

Reducing Hallucination in Legal RAG Chatbots: A Comparative Study of Deep Learning Retrieval Architectures

Project Group #43

COMPSCI 1090B: Data Science 2: Advanced Topics in Data Science

Final Report (MS4)

Background & Motivation

Legal hallucination is grounded in *Mata v. Avianca Airlines* (2023) - a documented case of legal hallucination with real-world consequences, where a New York attorney filed a brief containing fabricated case citations produced by ChatGPT. Stanford RegLab reports legal hallucination rates of 69-88% for general-purpose LLMs on specific legal queries, confirming this is a systemic issue rather than an isolated incident. Retrieval-augmented generation (RAG) is the dominant industry response: condition the model on documents pulled from a vetted corpus so the answer is grounded in real law rather than parametric guesswork.

The unanswered question for legal practice is which retrieval setup actually delivers grounded answers, and what in the pipeline drives the reduction in hallucination. Lawyers, vendors, and regulators need a quantitative answer because deployment decisions - whether to allow a tool in court filings, what disclosures to require, what residual risk to insure against - depend on the ceiling of retrieval-side mitigation. Our work targets U.S. federal appellate opinions from CourtListener (1,465,484 opinions) and measures, end-to-end, how alternative retrieval architectures translate into evidence grounding and downstream hallucination behavior.

Problem Statement

Research question. Which retrieval setup most improves evidence grounding and reduces contradiction and neutral-evidence failures in a legal RAG system built over U.S. federal appellate opinions?

Hypotheses. H1: a hybrid BM25 + BGE-M3 + cross-encoder pipeline achieves significantly higher Recall@10 than BM25 and BGE-M3 alone (paired bootstrap, $p < 0.05$). H2: architectures with higher Recall@10 produce significantly lower contradiction rates in downstream generation (normalized by claim count and per 1K tokens). H3: the hybrid achieves higher Recall@10 than BGE-M3 alone.

Target / output. For each (query, retrieval-configuration) pair, a structured judgment on a 3-class scale - FAITHFUL, PARTIAL, HALLUCINATED - produced by an LLM judge over the generator's answer with respect to the contexts the generator was shown. Aggregated, the target is a five-point hallucination-rate vector indexed by retrieval configuration, paired with retrieval Hit@k / MRR / NDCG@10 metrics so the H2 retrieval-quality $\rightarrow$ contradiction relationship can be tested directly.

Unit of observation. A single LePaRD (source_id, dest_id) citation pair after symmetric leakage cleaning and (source_id, dest_id) deduplication: 20,877 unique queries, each asking the model to reproduce a legally correct citation given the surrounding quote text.

Data & EDA

We use two public datasets joined by a custom semantic bridge.

CourtListener (58 GB, December 2025) provides the corpus: 1,465,484 federal appellate opinions across 13 circuits (ca9 15.6%, ca5 14.3%, ca4 12.2%, no dominance). Text length follows a power law on log-log axes: a flat plateau from ~50 to 8,000 characters and a six-order-of-magnitude tail to 100K+, with 8,465 opinions exceeding 100K characters. Citation density is heavily left-skewed. Filtering at 100 chars removes only 5,574 records (0.38%).

LePaRD (Mahari et al., ACL 2024) provides 4M (quote, source-cite, dest-cite) tuples scraped from federal appellate opinions. Citation frequency is itself long-tailed.

The pivotal EDA finding. A naive numeric-id audit between LePaRD and CourtListener yielded only 47,247 usable gold pairs (2.6%), threatening the evaluation budget. Investigation revealed: (1) no public CAP $\leftrightarrow$ CL ID crosswalk exists; (2) CourtListener already ingested the Caselaw Access Project under different IDs; (3) the legal-research community's canonical bridge is parallel citation strings (volume, reporter, page). We built a two-stage semantic bridge: stage 1 parses each LePaRD source_cite with eyecite and joins on CourtListener cluster_id (99.91% match); stage 2 validates the LePaRD quote appears in the destination opinion via rapidfuzz.partial_ratio $\geq$ 80. The bridge produces **2,429,533 verified pairs** (60.74%), a **51 $\times$ expansion** that makes downstream evaluation tractable.

Three additional EDA-driven decisions materially changed modeling: (i) symmetric leakage cleaning of query quotes and corpus chunks via RE2-linear-time stripping of case names and citations - without it, BM25 trivially recovers cited opinions and Hit@1 inflates from 2.5% to 18%+; (ii) cluster-level matching using cluster_id rather than opinion_id, because CourtListener represents the same legal text under multiple opinion IDs; (iii) (source_id, dest_id) deduplication. After all gates, the canonical evaluation universe is **20,877 unique queries** over a **7,813,273-chunk** cleaned corpus spanning **1,360,665 distinct clusters**.

Methods & Models

The pipeline is a five-stage funnel: cleaning → first-stage retrieval → fusion → cross-encoder reranking → answer generation and judging. Every stage is deterministic and provenance-locked (SHA-256 + git SHA + seed + DVC).

First-stage retrieval (two complementary baselines). BM25 (`bm25s`, $k_1=1.5$, $b=0.75$) indexes 7.8M chunks in 36 min, retrieves top-100 in 110 min single-threaded. BGE-M3 (BAAI/bge-m3, 1024-dim, `max_length=8192`, normalized cosine) is encoded across 4× NVIDIA L4 GPUs in DDP with cross-shard MaxP merge: 55 min retrieval (6.3 qps aggregate). Both operate on identical corpus and 20,877-query test set with `top_k=100` to keep comparisons apples-to-apples.

Hybrid fusion. Reciprocal Rank Fusion (Cormack et al. 2009) with $k=60$: fused score = $\sum_r 1 / (60 + \text{rank}_r(d))$. Parameter-light and unsupervised.

Cross-encoder reranking (three variants on the same RRF top-100 pool). *Concat*: hub `bge-reranker-v2-m3` with 2-chunk concat at `max_length=1024`. *MaxP*: hub encoder scoring up to 3 chunks per cluster independently, max-pooled. *Finetuned*: same architecture as concat but encoder fine-tuned on legal hard negatives.

Hard-negative mining and reranker fine-tuning (the project's "advanced method," meeting the AC209B not-covered-in-class requirement). For each query whose RRF top-100 contains gold, we sample 7 hard negatives from RRF ranks 2-100 with at most 2 chunks per cluster, producing 7,442 train and 391 val queries → 59,536 train pairs. We fine-tune `bge-reranker-v2-m3` with pointwise BCE at $\text{lr}=2\text{e-}5$, $\text{batch}=4 \times \text{grad-accum}=8 \times 4$ GPUs (effective batch 128), 2 epochs (930 steps), `max_length=1024`, fp16, on 4× L4 DDP - 5h38min wallclock (~22 GPU-hours). Hard-negative mining follows BAAI's official guidance, Karpukhin et al. (DPR 2020), and the 4Huiter SoICT 2024 winning recipe (+23% MRR on Vietnamese legal). Cross-encoder rerankers and the BGE family are not part of the standard CS109b material.

PARADE-Transformer aggregator (Li et al., SIGIR 2020 / TOIS 2023). On top of the fine-tuned encoder, a 2-layer 8-head Transformer encodes up to 8 per-passage [CLS] vectors (frozen encoder; ~1M aggregator params), 3 epochs, $\text{lr}=5\text{e-}5$. PARADE addresses the long-document scoring problem visible in EDA Figure 2.

RAG generation. For each query and each of five ablations (`none`, `bm25`, `bge_m3`, `rrf`, `reranker_finnetuned`) we generate one Qwen2.5-7B-Instruct answer grounded on top-5 retrieved cluster contexts (or no context for `none`). Each ablation runs as its own SLURM job, query-sharded across 4× L4. Total ~50 GPU-hours for $5 \times 20,877 = 104,385$ generations.

LLM-as-judge hallucination evaluation. Each generation is judged by `gpt-4o-mini` against the contexts shown, returning `FAITHFUL` / `PARTIAL` / `HALLUCINATED`. We budget ~\$53 of \$65 OpenAI credit; sample sizes give 95% CIs of $\pm 0.86\%$ to $\pm 1.96\%$.

Splits and reproducibility. LePaRD provides only test (45K → cleaned 20,877) and val (1,692) splits; reranker training uses only queries whose RRF top-100 contained gold. Cluster-level matching guarantees zero leakage between training and test (verified). Every artifact ships with

.summary.json carrying SHA-256 + git SHA + seed; the canonical artifact final_summary.json is itself hashed (43eec4d3023f9485...).

Results & Interpretation

Retrieval ladder (n=20,877)

Retriever	Hit@1	Hit@5	Hit@10	Hit@100	MRR	NDCG@10
BM25	0.0251	0.1027	0.1459	0.3332	0.0642	0.0783
BGE-M3	0.0244	0.0631	0.0862	0.2216	0.0457	0.0516
RRF	0.0391	0.1081	0.1557	0.3752	0.0769	0.0894
reranker_concat (hub)	0.0284	0.0787	0.1098	0.3752	0.0575	0.0639
reranker_maxp (hub)	0.0426	0.1039	0.1470	0.3752	0.0778	0.0881
reranker_finetuned	0.3069	0.3516	0.3598	0.3752	0.3275	0.3349

Table 1: Aggregate retrieval metrics. Reranker variants share RRF's Hit@100 ceiling because reranking only re-orders the candidate pool.

Comparison to MS3 baseline. MS3 reported BM25 marginally beating BGE-M3 (Hit@10: 0.146 vs 0.086), leaving open whether this was a sparse-vs-dense fundamentals gap or a domain-adaptation gap. MS4 answers it. Hub bge-reranker-v2-m3 with concat (Hit@10=0.110) is worse than RRF - out-of-domain reranking actively degrades legal retrieval. Hub MaxP recovers most of the concat truncation loss (0.147, comparable to RRF) but cannot exceed lexical fusion. The fine-tuned reranker, identical architecture and identical RRF candidate pool, lifts Hit@1 from 0.0284 to 0.3069 - a **+980% relative gain, 12.2× above BM25's Hit@1 ceiling**. The MS3 negative result was a domain-adaptation gap, recoverable with the right training data.

Per-query paired analysis. RRF wins only 17.1% of queries against BM25 on strict per-query rank but beats it on aggregate Hit@10 - the gain comes from cases both retrievers find gold and RRF nudges it across a top-k cutoff. Against the fine-tuned reranker: BM25 wins 6.6%, RRF 2.0%, hub-concat 1.1% (a 30.6× ratio in the reranker's favor - the cleanest measurement of the legal domain-adaptation gap available). Tie rates rise monotonically as variants strengthen (59.4% → 66.0%), reflecting that ~60-66% of queries are in the both-miss zone bounded by RRF's Hit@100=0.375 ceiling.

Stratified evaluation (HEAD / TORSO / TAIL by gold-cluster citation frequency)

Retriever	Overall	HEAD	TORSO	TAIL
BM25	0.1459	0.1318	0.1878	0.2393
BGE-M3	0.0862	0.0794	0.1061	0.1319
RRF	0.1557	0.1427	0.1946	0.2403
reranker_concat (hub)	0.1098	0.0914	0.1719	0.2065
reranker_maxp (hub)	0.1470	0.1227	0.2282	0.2751

Retriever	Overall	HEAD	TORSO	TAIL
reranker_finetuned	0.3598	0.3596	0.3694	0.3292

Table 2: Hit@10 stratified by gold-cluster citation frequency. HEAD = top tertile (16,507 queries, 486 most-cited clusters, $\text{freq} \in [21, 2596]$); TAIL = bottom tertile (978 queries, 487 rarest clusters, $\text{freq} \in [1, 5]$).

The five hub variants exhibit a **counter-intuitive inverted long-tail**: TAIL Hit@10 exceeds HEAD by 1.66 $\times$ -2.26 $\times$. Rare precedents have lexically-distinctive citation contexts; frequent precedents (constitutional canon) carry generic contexts that match many candidates. Domain fine-tuning *flips* the pattern (HEAD $\approx$ TORSO > TAIL) because hard negatives were 79% HEAD-dominated by query volume - the model traded TAIL specialization for HEAD discrimination. This contradicts the dominant "long-tail problem" framing (AusLaw December 2025): the real problem in legal RAG is the *high-frequency precedent* problem, exactly the cases (constitutional canon, *Chevron*, *Bell v. Hood*) where citation faithfulness matters most.

RAG hallucination (live counts from `judgments.jsonl`)

Ablation	n_judged	95% CI	Hit@10	Faithful	Partial	Hallucinated
none	12,977	$\pm 0.86\%$	-	0.0006	0.0000	0.9994
bge_m3	7,101	$\pm 1.16\%$	0.0862	0.0907	0.2694	0.6399
bm25	6,929	$\pm 1.18\%$	0.1459	0.0915	0.2793	0.6292
rrf	7,017	$\pm 1.17\%$	0.1557	0.0949	0.3024	0.6027
reranker	2,500	$\pm 1.96\%$	0.3598	0.1184	0.3240	0.5576

Table 3: 5-ablation hallucination matrix.

Three findings. First, `no_rag` is 99.94% hallucinated - Qwen2.5-7B has near-zero parametric knowledge of the LePaRD citation graph. Second, hallucination rate is **monotonically inverse to Hit@10**; the Pearson correlation across the four RAG ablations is $r = -0.9624$ ($r^2 = 92.6\%$). Third, the fine-tuned reranker (Hit@10 = 0.3598, +131% over RRF) reduces hallucination by 4.5pp absolute / 7.5% relative and lifts faithfulness by 25% - meaningful but not proportional, evidence of diminishing returns.

The total reduction from `no_rag` (0.9994) to `reranker_rag` (0.5576) is **44.2 percentage points** - RAG roughly halves hallucination. Stanford RegLab's bare-LLM baseline of 69-88% is *lower* than our 99.94% no-RAG floor because we score on a stricter cluster-level criterion; their retrieval-grounded analogue would land in our 56-64% RAG band.

Critical project finding: the legal-RAG hallucination floor is approximately 56% at current SOTA retrieval. Even with a +980% Hit@1 gain from domain fine-tuning, more than half of legal RAG answers remain hallucinated. This is the central empirical contribution: a quantified retrieval-ceiling $\rightarrow$ hallucination-floor relationship complementing Stanford RegLab's bare-LLM baselines.

Error analysis

The 60-66% paired-tie zone - predominantly both-miss queries given the $\text{Hit@100} = 0.375$ ceiling - defines a hallucination floor retrieval cannot lower. Within that zone the model receives top-5 contexts not containing gold, and Qwen2.5-7B degrades from grounded citation to confident fabrication. The slope between rrf ($\text{Hit@10}=0.156$, $\text{halluc}=0.603$) and reranker (0.360, 0.558) is $\sim 2\text{pp}$ hallucination per 0.1 Hit@10 - diminishing returns. The HEAD weakness in the hub regime ($\text{Hit@1} \approx 0.02\text{-}0.03$ on 79% of queries) is invisible in aggregate metrics and only surfaces in stratified evaluation.

Conclusions & Discussion

We can claim three things with publication-grade support. **(1)** Retrieval quality is the dominant lever for RAG hallucination on legal citation QA: $r = -0.9624$ across four configurations spanning Hit@10 from 0.086 to 0.360 yields $r^2 = 92.6\%$. **(2)** Domain fine-tuning of a cross-encoder reranker on 7,442 legal hard negatives is the precise mechanism that closes the legal-domain gap (+980% Hit@1 over the same hub checkpoint, +1,120% over BM25), confirming the MS3 result was a domain-adaptation gap, not a sparse-vs-dense gap. **(3)** The legal-RAG hallucination floor at SOTA retrieval is $\sim 56\%$; doubling Hit@10 from 0.36 would not push hallucination below $\sim 30\%$ in our linear regime.

We **cannot** claim the floor generalizes beyond LePaRD-style federal appellate citation tasks: our judge is single-model `gpt-4o-mini` ($\sim 80\%$ accuracy on FaithBench), our generator is one specific 7B model. Cross-judge agreement would tighten absolute rates; *relative* differences across ablations are robust because judge bias is constant.

Limitations. (i) Reranker $n=2,500$ vs others $n=6,929\text{-}12,977$ due to budget; CIs still non-overlapping with the headline gap. (ii) LLM-as-judge has documented limits. (iii) Cluster-level matching is generous (any cluster opinion counts) and strict (chunk-level partials don't). (iv) No temporal split. (v) PARADE training completed but inference at scale was descoped; MaxP and finetuned variants were sufficient.

Future work. The +980% retrieval gain $\rightarrow$ 25% relative faithfulness gain ratio suggests the next lever is generation-side: constrained decoding to citation tokens, citation-aware abstention prompting, knowledge-graph-aligned hallucination detection (HalluGraph 2025), Finetune-RAG-style supervised refusal. Beyond that, graded relevance labels would unlock NDCG sensitivity, and a temporal split would harden deployment claims.

Broader Impact

Who benefits. Practicing lawyers and judges weighing AI-assisted research; vendors needing empirical SOTA bounds; courts and bar associations writing disclosure rules; insurers pricing malpractice on AI work product. A 56% floor at SOTA retrieval is a concrete number supporting Stanford RegLab's "careful, supervised integration" recommendation - the residual rate is too high for unsupervised deployment in citation-critical work.

Risks. A system optimized only on aggregate metrics will quietly underperform on HEAD queries (the most-cited precedents) - exactly the constitutional-canon cases where a missed or fabricated cite is most consequential. False confidence is an active harm: a hallucinated cite that "looks right" can pass review and reach a filing, as the New York case demonstrated. Our retrieval-floor finding should be paired with a stratified disclosure norm - Hit@k by precedent-frequency tertile, not just aggregate.

Connections. The work intersects three live AI-governance debates: (a) whether retrieval is sufficient mitigation for high-stakes hallucination (necessary, not sufficient); (b) whether domain fine-tuning of open models is competitive with frontier closed APIs on specialized tasks (yes, with 22 GPU-hours of L4 time on 7,442 legal pairs); (c) whether long-tail evaluation should be standard rather than appendix (yes - the inverted long-tail flips the deployment story).

References

Datasets. LePaRD (Mahari et al., ACL 2024). CourtListener bulk dump, Free Law Project, December 2025.

Pretrained models. BAAI/bge-m3; BAAI/bge-reranker-v2-m3; Qwen/Qwen2.5-7B-Instruct; OpenAI gpt-4o-mini (judge).

Methods. Cormack et al., *Reciprocal Rank Fusion*, SIGIR 2009. Karpukhin et al., *Dense Passage Retrieval*, EMNLP 2020. Li et al., *PARADE*, ACM TOIS 2023. Stanford RegLab, *Hallucinating Law*, 2024.

Code repository.

https://github.com/ltphongssvn/cs1090b_HallucinationLegalRAGChatbots

Appendix A - Industry research papers cited in the notebook

Hyperlinks to the industry/academic sources surfaced through web search during MS3/MS4 cell interpretation, grouped by where they appear in the analysis.

Cross-encoder reranking, hard-negative mining, and domain fine-tuning (Cells 17b, 17c, 18).

- *Cross-Encoder Reranking* (overview), Emergent Mind, <https://www.emergentmind.com/topics/cross-encoder-reranking-9dd25a04-77c6-4f44-807d-cb5f2256901b>
- *Cross-Encoder Rerankers* (overview), Emergent Mind, <https://www.emergentmind.com/topics/cross-encoder-rerankers>
- *Cross-Encoder Reranking* (mechanistic interpretability of BM25-like circuits), Emergent Mind, <https://www.emergentmind.com/topics/cross-encoder-reranking>
- Pezzuti et al., *Exploring the Effectiveness of Multi-stage Fine-tuning for Cross-encoder Re-rankers*, arXiv 2503.22672, <https://arxiv.org/html/2503.22672v1>
- *Hard Negatives, Hard Lessons: Revisiting Training Data Quality for Robust Information Retrieval with LLMs*, arXiv 2505.16967, <https://arxiv.org/pdf/2505.16967>
- *Improving Code Search with Hard Negative Sampling Based on Fine-tuning*, arXiv 2305.04508, <https://arxiv.org/pdf/2305.04508>
- *Optimizing Legal Document Retrieval in Vietnamese with Semi-Hard Negative Mining* (4Huter SoICT 2024), arXiv 2507.14619, <https://arxiv.org/pdf/2507.14619>
- *Advanced RAG Retrieval: Cross-Encoders & Reranking*, Towards Data Science, <https://towardsdatascience.com/advanced-rag-retrieval-cross-encoders-reranking/>
- *Fine-tuning Cross-Encoders for Re-ranking*, Dev-kit, <https://dev-kit.io/blog/ai/fine-tuning-cross-encoders-re-ranking>
- *Adaptive Retrieval Reranking - Enterprise RAG Ranking Failures*, generation RAG, <https://ragaboutit.com/adaptive-retrieval-reranking-how-to-implement-cross-encoder-models-to-fix-enterprise-rag-ranking-failures/>
- *In Defense of Cross-Encoders for Zero-Shot Retrieval*, arXiv 2212.06121, <https://arxiv.org/pdf/2212.06121>
- *A Thorough Comparison of Cross-Encoders and LLMs for Reranking SPLADE*, arXiv 2403.10407, <https://arxiv.org/html/2403.10407v1>
- *Cross-Encoder Reranking: The Low-Cost Fix for RAG Misses*, blog.thegenairevolution, <https://blog.thegenairevolution.com/article/cross-encoder-reranking-the-low-cost-fix-for-rag-misses>
- *Sparse vs Dense Retrieval for RAG*, ML Journey, <https://mljourney.com/sparse-vs-dense-retrieval-for-rag-bm25-embeddings-and-hybrid-search/>
- *Passage Retrieval of Polish Texts Using OKAPI BM25 and an Ensemble of Cross-Encoders*, arXiv 2410.04620, <https://arxiv.org/pdf/2410.04620>

PARADE and long-document retrieval (Cell 17d).

- Li et al., *PARADE: Passage Representation Aggregation for Document Reranking*, arXiv 2008.09093, <https://arxiv.org/abs/2008.09093> ; ACM TOIS final at <https://dl.acm.org/doi/10.1145/3600088> ; PDF at <https://arxiv.org/pdf/2008.09093> ; HTML at <https://ar5iv.labs.arxiv.org/html/2008.09093> ; Glasgow eprint <https://eprints.gla.ac.uk/298257/2/298257.pdf>
- *PARADE official code repository*, GitHub canjiali/PARADE, <https://github.com/canjiali/PARADE>
- *A Survey of Long-Document Retrieval in the PLM and LLM Era*, arXiv 2509.07759, <https://arxiv.org/pdf/2509.07759> ; HTML <https://arxiv.org/html/2509.07759v1>
- *Long-Document Retrieval Overview*, Emergent Mind, <https://www.emergentmind.com/topics/long-document-retrieval-ldr>
- *Long Document Re-ranking with Modular Re-ranker*, arXiv 2205.04275, <https://arxiv.org/pdf/2205.04275>
- *PARM: A Paragraph Aggregation Retrieval Model*, arXiv 2201.01614, <https://arxiv.org/pdf/2201.01614>
- *DoSSIER@COLIEE 2021: Dense retrieval and summarization-based re-ranking for case law*, arXiv 2108.03937, <https://arxiv.org/pdf/2108.03937>
- *Neural reranking for UK statutory retrieval (UK-StatuteCorpus)*, Springer, <https://link.springer.com/article/10.1007/s10506-025-09501-6>

Stratified / long-tail legal retrieval (Cell 18b).

- *Legal citation prediction with LLMs - AusLaw citation benchmark*, Springer, <https://link.springer.com/article/10.1007/s10506-026-09506-9>
- Mao, *Exploring Selective Retrieval-Augmentation for Long-Tail Legal Text Classification*, arXiv 2508.19997, <https://arxiv.org/abs/2508.19997> ; PDF <https://arxiv.org/pdf/2508.19997> ; HTML <https://arxiv.org/html/2508.19997>
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- *A Lawyer's Quick Guide To Long-Tail Keywords*, Justia onward blog, <https://onward.justia.com/a-lawyers-quick-guide-to-long-tail-keywords/>
- *Bridging the Long-Tail Gap: Robust RAG via Multi-Stage Paraphrase Infusion*, arXiv 2604.22261, <https://arxiv.org/html/2604.22261>
- *RPDR: A Round-trip Prediction-Based Data Augmentation Framework for Long-Tail QA*, arXiv 2602.17366, <https://arxiv.org/pdf/2602.17366>
- *Answer Retrieval in Legal Community Question Answering (LegalQA)*, arXiv 2401.04852, <https://arxiv.org/pdf/2401.04852>

Paired-comparison and metric-disagreement evaluation (Cell 19).

- Diaz, *Offline Retrieval Evaluation Without Evaluation Metrics (RPP)*, arXiv 2204.11400, <https://arxiv.org/pdf/2204.11400>
- *Reliable Evaluation Protocol for Low-Precision Retrieval (HPS + tie-aware metrics)*, arXiv 2508.03306, <https://arxiv.org/pdf/2508.03306>
- *Correlation and Prediction of Evaluation Metrics in Information Retrieval*, arXiv 1802.00323, <https://arxiv.org/pdf/1802.00323>
- *How to Evaluate Retrieval Quality in RAG Pipelines: Precision@k, Recall@k, F1@k*, Towards Data Science, <https://towardsdatascience.com/how-to-evaluate-retrieval-quality-in-rag-pipelines-precisionk-recallk-and-f1k/>

- *Evaluating the Retrieval Component in RAG*, apxml.com, <https://apxml.com/courses/getting-started-rag/chapter-6-evaluating-improving-rag-systems/evaluating-retrieval>
- *Evaluating the Retrieval Pipeline*, Medium, <https://medium.com/@debraj.bhattacharya83/evaluating-the-retrieval-pipeline-key-to-a-high-performing-rag-chatbot-c9e1d0421b72>

RAG hallucination, ablation design, and judgment infrastructure (Cells 20, 21).

- *Beyond Component Strength: Synergistic Integration and Adaptive Calibration in Multi-Agent RAG*, arXiv 2511.21729, <https://arxiv.org/pdf/2511.21729>
- *Retrieval Improvements Do Not Guarantee Better Answers: A Study of RAG for AI Policy QA*, arXiv 2603.24580, <https://arxiv.org/pdf/2603.24580>
- *Stable-RAG: Mitigating Retrieval-Permutation-Induced Hallucinations*, arXiv 2601.02993, <https://arxiv.org/html/2601.02993>
- *Enhancing Legal LLMs through Metadata-Enriched RAG Pipelines and DPO*, arXiv 2603.19251, <https://arxiv.org/pdf/2603.19251>
- *Towards Reliable Retrieval in RAG Systems for Large Legal Datasets*, arXiv 2510.06999, <https://arxiv.org/pdf/2510.06999>
- *MEGA-RAG: multi-evidence guided answer refinement* (PMC, public health RAG ablations), <https://pmc.ncbi.nlm.nih.gov/articles/PMC12540348/>
- *Towards Lighter and Robust Evaluation for Retrieval Augmented Generation*, arXiv 2503.16161, <https://arxiv.org/pdf/2503.16161>
- *Stanford RegLab - Hallucinating Law: Legal Mistakes with Large Language Models are Pervasive*, Stanford Law School, <https://law.stanford.edu/2024/01/11/hallucinating-law-legal-mistakes-with-large-language-models-are-pervasive/>
- *Benchmarking LLM Faithfulness in RAG with Evolving Leaderboards (FaithJudge)*, arXiv 2505.04847, <https://arxiv.org/html/2505.04847v1> ; PDF <https://arxiv.org/pdf/2505.04847> ; ACL Anthology EMNLP Industry 2025 <https://aclanthology.org/2025.emnlp-industry.54.pdf>
- *FaithBench: A Diverse Hallucination Benchmark for Summarization by Modern LLMs*, arXiv 2410.13210, <https://arxiv.org/pdf/2410.13210>

Pearson r, RAG hallucination correlation, retrieval as primary driver (Cells 22, 23).

- *HalluGraph: Auditable Hallucination Detection for Legal RAG via Knowledge Graph Alignment*, arXiv 2512.01659, <https://arxiv.org/pdf/2512.01659>
- *Retrieval Augmentation Reduces Hallucination in Conversation*, ACL Findings 2021, <https://aclanthology.org/2021.findings-emnlp.320.pdf>
- *Retrieval-augmented generation elevates local LLM quality in radiology contrast media consultation*, PMC, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12223273/>
- *NoMIRACL: Knowing When You Don't Know - Multilingual Relevance Assessment for Robust RAG*, arXiv 2312.11361, <https://arxiv.org/pdf/2312.11361>
- *ReDeEP: Detecting Hallucination in RAG via Mechanistic Interpretability* (Pearson r 0.41 / 0.46 baseline), arXiv 2410.11414, <https://arxiv.org/html/2410.11414v2>
- *LUMINA: Detecting Hallucinations in RAG with Context-Knowledge Signals*, arXiv 2509.21875, <https://arxiv.org/pdf/2509.21875>
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